

MATH 130: 9/17 WORKSHEET

Last time we talked about the language of *propositional logic*. Today we want to talk about its *semantics*—how do we explain the meaning of the connectives? We will do with the device of a *truth table*, invented by Austrian philosopher Ludwig Wittgenstein. This device is an exhaustive listing of how a symbol behaves for any possible combination of inputs.

Our semantics will be *truth-functional*, meaning that to know whether a statement is true or false you only have to know the truth values of the inputs.

Conjunction (and).

P	Q	$P \wedge Q$
T	T	
T	F	
F	T	
F	F	

Conditional (if-then).

P	Q	$P \Rightarrow Q$
T	T	
T	F	
F	T	
F	F	

Disjunction (or).

P	Q	$P \vee Q$
T	T	
T	F	
F	T	
F	F	

Bi-conditional (iff).

P	Q	$P \Leftrightarrow Q$
T	T	
T	F	
F	T	
F	F	

Negation (not).

P	$\neg P$
T	
F	

You can also use truth tables to analyze more complicated sentences with multiple connectives.